PROGREE – DATA SCIENCE INTERNSHIP

Final Tasks Submission

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1. Submission Overview

This archive contains the completed projects prepared as part of the Progree Data Science Internship.

The submission includes three main technical tasks, along with the optional internship acceptance task.

2. Submitted Tasks

Task 1 – Internship Acceptance (Optional)

Documentation related to the internship acceptance and initial participation.

Task 2 – Data Cleaning

A practical project focused on data quality assessment, data cleaning, and preparing structured datasets for further analysis.

Included files:

- Jupyter Notebook
- Dataset files
- Project outputs

Task 3 – Customer Segmentation

A customer segmentation project using data analysis and machine learning techniques to identify distinct customer groups.

Included files:

- Jupyter Notebook
- Dataset files

- Project outputs

Task 4 – Bike Rental Demand Forecasting & Model Interpretability

A time-series forecasting project using SARIMAX, supported by SHAP-based feature importance analysis through a supplementary Random Forest model.

Included files:

- Bike_Rental_Forecasting.ipynb
- Bike_Rental_Forecasting_Whitepaper_Final.pdf
- Forecasting and evaluation outputs
- README.md

3. Technologies Used

- Python
- Pandas & NumPy
- Matplotlib & Seaborn
- Scikit-learn
- Statsmodels
- SHAP
- Jupyter Notebook

4. Submission Structure

The archive is organized into separate folders for each task, containing the relevant notebooks, datasets, outputs, and documentation.

5. Acknowledgment

I would like to thank Progree for providing this learning opportunity and supporting practical experience in Data Science.

Submitted as part of the Progree Data Science Internship.